

Objective:

Engineering student who treats silicon as a canvas and code as architecture, driven to design embedded and VLSI systems that translate imagination into measurable impact. Seeking an environment where deep technical rigor, ownership, and bold thinking converge to build technology that stands the test of time.

Area of Interest:

- **Embedded Systems**(RTOS, Development, Automation, IOT)
- **Renewable Energy** (Smart grid, BMS, Smart energy utilization)
- **Automation** (Sensor interface, Automation, Circuit analysis)

Education:

B.E in Electrical and Electronics Engineering | Dr.Mahalingam College of Engineering and Technology, Pollachi.

October, 2027 | CGPA: 7.9

HSC | Equitas Gurukul Matric Higher Secondary School, Dindigul.

April, 2023 | Percentage: 71%

Skills:

Technical Skills:

- **Programming Languages:** C, Python, Java, Verilog.
- **Robotics, Embedded and IOT Tools:** Linux, Firebase, Arduino IDE, GIT
- **Bords known to work:** ESP(32 ,826), Raspberry pi (3,pico,zero), Arduino(UNO,NANO,MEGA).
- **AI Automation:** n8n.
- **Circuit Simulation:** LT Spice, Matlab.
- **PCB Design:**Kicad.
- **AI tool:**Antigravity, Claude code, Cursor and Etc.

Soft Skills:

- **Communication:** Conveys technical ideas clearly and confidently.
- **Adaptability:** Thrives in dynamic and evolving situations.
- **Teamwork:** Collaborates effectively in diverse environments
- **Cross-Domain Integration:** Bridges hardware, software, and system-level thinking effectively.
- **Calm Technical Judgment:**Rational under pressure

Projects:

Smart Agro-Environmental Monitoring System (SAEMS)

May 2024

- An ESP32-based AgroClimate IntelliSense System delivers real-time soil and atmospheric insights, enabling precision farming, optimized irrigation, and data-driven agricultural decision-making.

Automated water management system(SIH)

Oct 2025

- An Automated Water Management System intelligently monitors and controls water distribution in real time, minimizing wastage and maximizing efficiency through smart sensing and automation.

Tifan (Seed driller / Planter)

Apr 2026

- A Tifan (Seed Driller/Planter) is a precision agricultural implement designed to sow seeds uniformly at optimal depth and spacing, improving crop yield while reducing manual labor and seed wastage.

Inventory Management (Real life application)

June 2026

- A role-based inventory management system for tracking equipment, vendors, products, utilities, and audit assets with advanced search and reporting features.

Adaptive Smart Laboratory Management Framework using RFID and Web Technologies

Ongoing

- An RFID-enabled web application for managing laboratory assets, student attendance, experiments, and equipment through a centralized platform.

Internships:

BOSCH RETHROX [Pollachi]

Jun 2025

- Completed an internship at Bosch Rexroth AG, gaining practical exposure to industrial automation, hydraulics, and motion control systems.
- Assisted in understanding real-time drive and control applications, strengthening my knowledge in advanced manufacturing technologies and smart industrial solutions.

IITM IEAC(Industrial Energy Assessment cell)

Jun 2026

- Developed an Inventory Management Application for IIT Madras KISEM.
- Contributed to energy audit workflow optimization and industrial data analysis.

Siemens (Virtual : Industry readiness program)

On going

- Completed Siemens Industrial Simulation training with hands-on experience in engineering simulation tools and workflows.
- Applied simulation techniques to analyze system performance and strengthen problem-solving skills in engineering applications.

Certifications:

- NPTEL : Designing through Verilog programming .
- Participated in Advance tidal technology by department of Marine Engineering Indian institute technology.
- Workshop in Programming Rasberry pi using rednode by Top Engineers (IITM)
- NPTEL : Leadership and team effectiveness.

Achievements:

- Won best prototyping award in **TIFAN by SAE - 2026**
- Won 2nd price in Technical paper presentation at Bannari Amman institute of technology (banari ,erode).
- Won 2nd price in Cricuit verse by EEE department of Dr.Mahalingam collage of engineering and technology (Pollachi , coimbatore).
- 2 price in Code clash and IOT hackathon by EEE department of Dr.Mahalingam collage of engineering and technology (Pollachi , coimbatore).